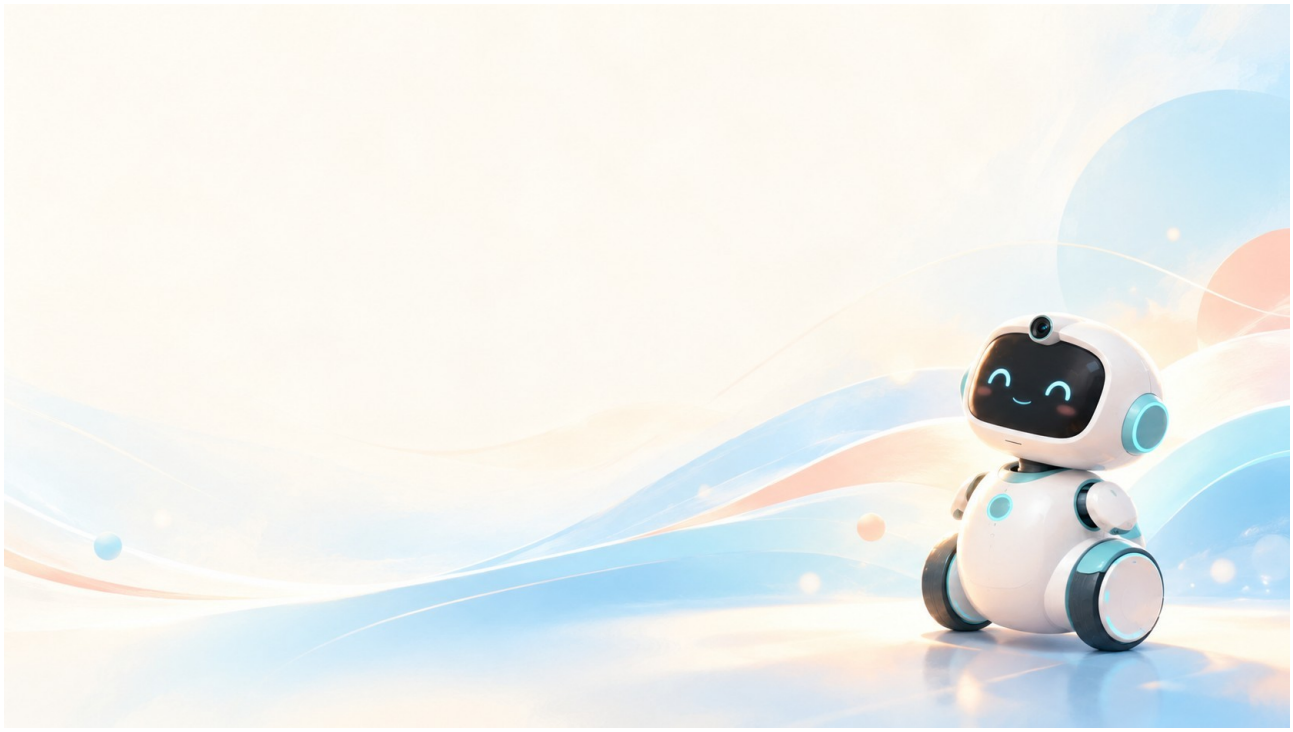
Data Analysis for the Development of Companion Robots for Older Adults

Interaction experience, continued use and the Guardian proposal



AI-generated concept illustration. It is not a commercial product, a research photograph or a real incident.

Purpose of the analysis

This project uses conversation records, user questionnaires and published studies to identify improvements that a research and development team could test. The main findings support attention to individual experience, continued use and reliable support workflows.

Guardian is the English name used here for the proposed companion robot concept. Its functions and test plan are proposals, not completed experiments or validated product capabilities.

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Student Name: Wong Chei Oi (Alice)

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How to read this report

The report starts with the HRI CUES data, then uses separate product studies to examine everyday use and limitations. It ends with design suggestions and a proposed test plan. Figure numbers follow the Notebook, so their numerical order differs from the reading order.

The supplementary studies involve different participants, settings, time periods and measures. They are interpreted separately and are not combined into a product ranking.

Summary and introduction

Summary

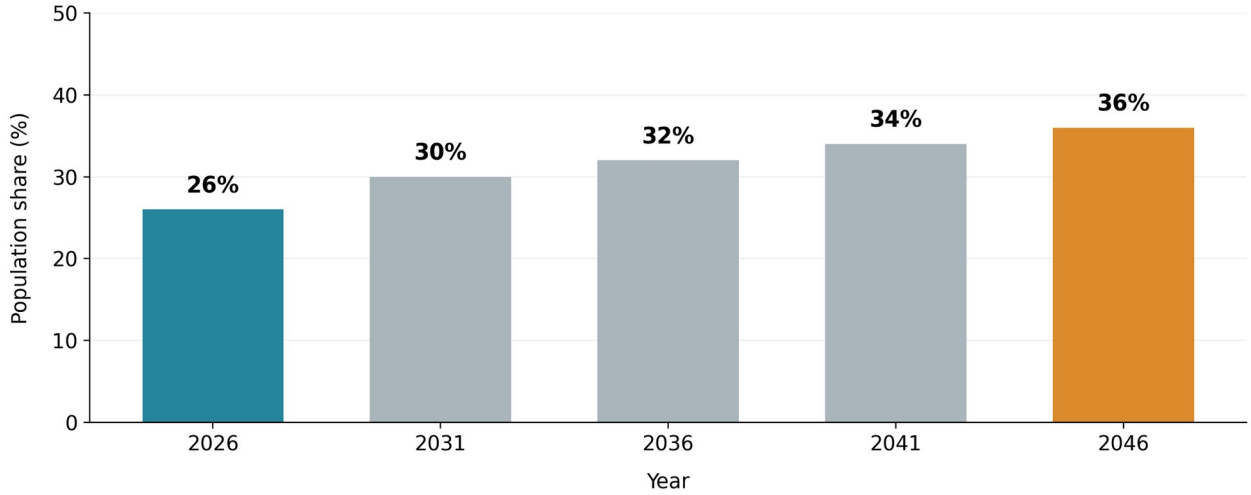
Population ageing provides a reason to study additional forms of support for older adults. This project analyses 590 dialogue turns and questionnaire records from 25 participants in the HRI CUES dataset. Experiences vary across people and turns, and the association between user self-reports and overall observer ratings is weak in the exploratory comparison. Published product studies add evidence about activities, short-term changes, continued use and functional availability. Together, these findings suggest improvements that should be tested locally before wider use.

1 Introduction

The Hong Kong Census and Statistics Department's 2023 projection shows the population aged 65 and over rising from 26% in 2026 to 36% in 2046, excluding foreign domestic helpers (Figure 7). This is a projection of population structure. It does not establish demand for a particular robot.

The main research question is: what do conversation records and ratings reveal about interaction experience, and which improvements should a product team examine next? Further questions concern continued use and the reliability of support functions. My role is to analyse the evidence and help a research and development team identify testable improvements.

Figure 7 Projected share of the Hong Kong population aged 65 and over



Note. All five values are projections published in 2023, excluding foreign domestic helpers. Source: Census and Statistics Department (2023), Appendix B, Table B1.

2 Data and methods

HRI CUES stands for Human-Robot Interaction Conversational User Enjoyment Scale. Observers use it to assess user enjoyment during robot conversations. The user questionnaire provides a separate self-reported perspective (Irfan et al., 2024, 2026). The main data come from brief Swedish workshop interactions, rather than the five supplementary product studies.

Data source	What a record represents	Use in this project
HRI CUES Exchanges	One dialogue turn; 590 turns from 25 people	Turn counts, duration, observer scores and score spread
HRI CUES Overall	One participant; 25 questionnaire records	User items and overall observer ratings
Nine supplementary CSVs	61 published aggregate records across separate sources	Population background, product studies and fraud context

Data preparation

The Notebook reads Exchanges with a semicolon separator and Overall with a comma separator. It checks column types, missing values and duplicate records. There are no duplicate participant-turn keys. P21 has no satisfaction answer; that value remains missing. The three-positive-item comparison therefore uses 24 complete participants.

Start and end timestamps are converted to seconds and subtracted. For example, 00:00:09.270 minus 00:00:00.000 gives 9.27 seconds. A turn contains robot and user responses, so its duration is not system response latency. The three observer scores are averaged within each turn. Overall observer ratings are taken separately from Overall.

Analysis process

Check the data, calculate turn-level and participant-level summaries, then use charts to answer the research questions. The analysis uses counts, means, distributions, a descriptive linear regression line and Spearman rank correlation. The mean of satisfaction, fun and interest is an exploratory three-item measure; it is not a validated replacement for the published scale. Strangeness and the supplied composite are excluded from that comparison.

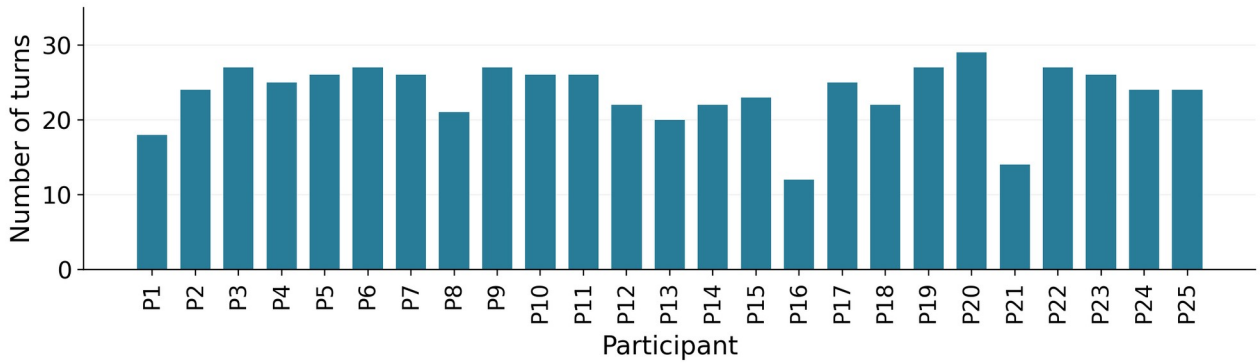
Repeated turns from the same participant are not independent people. Supplementary summaries are analysed within their own studies, without pooling samples or treating their 61 rows as 61 participants.

3 Analysis results

Dialogue turns per participant

Each bar represents one participant's dialogue turns. Counts range from 12 to 29, averaging 23.6 turns (Figure 1). Unequal counts describe the available records. A workshop time limit means more turns do not establish greater enjoyment or willingness to continue.

Figure 1 Dialogue turns per participant

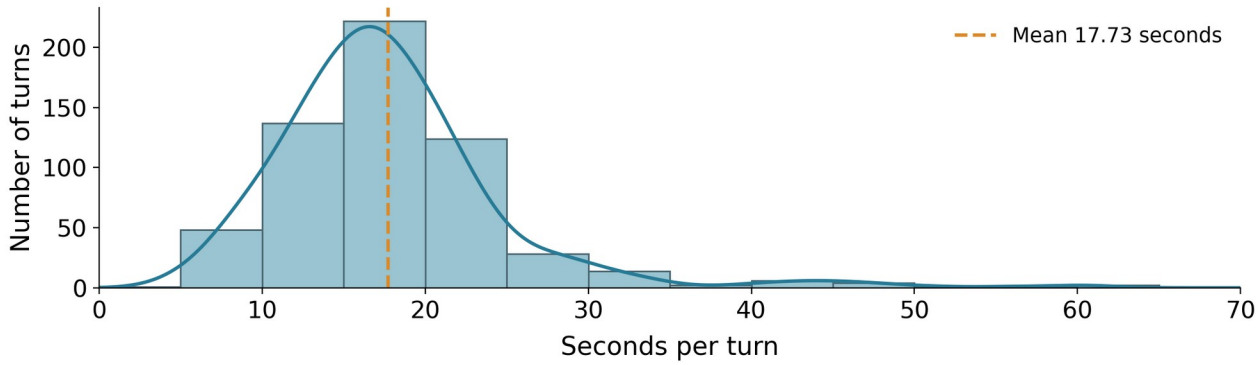


Note. 590 turns from 25 participants. Source: Irfan et al. (2024), Exchanges; calculations in the Notebook.

Distribution of turn duration

Bars count turns in five-second intervals (Figure 2). Mean duration is 17.73 seconds and the median is 17 seconds. The longer right tail shows a few longer turns. The smooth curve describes the sample distribution; the dashed line marks the mean. Duration includes both speakers and does not measure system latency or enjoyment.

Figure 2 Distribution of dialogue turn duration



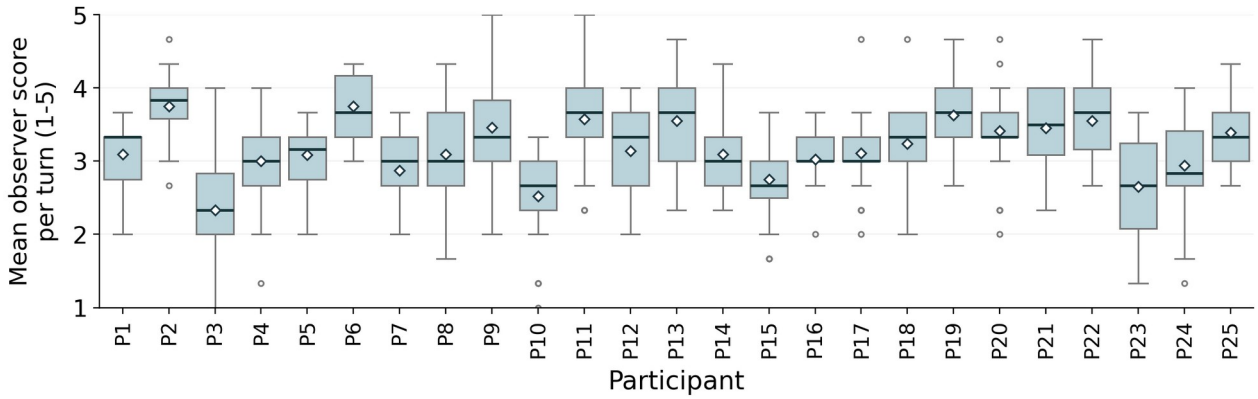
Note. The smooth curve is a kernel density estimate, not an assumed normal distribution. Source: Irfan et al. (2024), Exchanges; calculations in the Notebook.

Individual experience and rating distributions

Enjoyment varies across dialogue turns

Figure 3 shows variation within each person's turns and differences across participants. Each turn score averages three observer ratings. The box shows the middle 50%, the line is the median, and the white diamond is the mean. This variation supports collecting individual feedback rather than assuming one interaction style suits everyone.

Figure 3 Observed enjoyment across dialogue turns

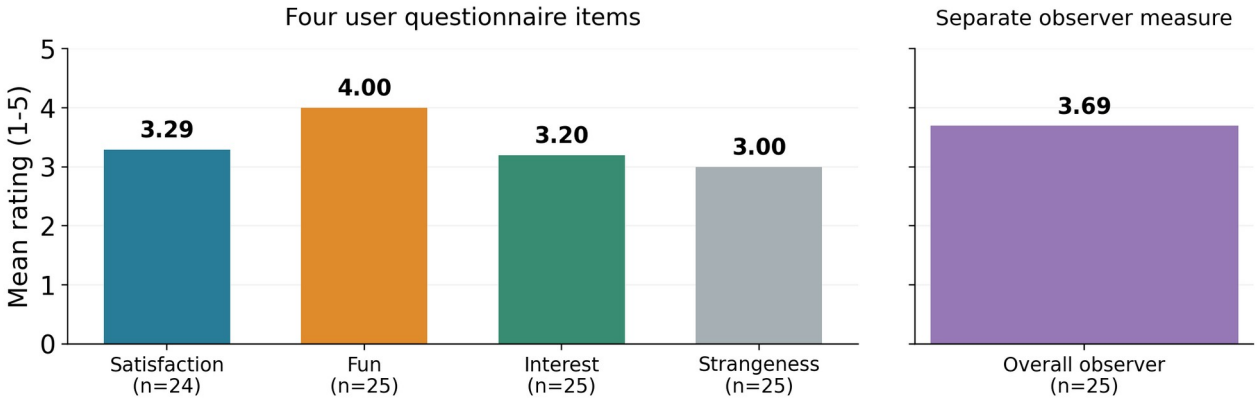


Note. Whiskers follow the 1.5-IQR rule. Circles beyond them are outliers, not necessarily recording errors. Turns within a person are repeated observations. Source: Irfan et al. (2024), Exchanges.

User questionnaire and observer ratings

In Figure 4, mean user ratings are 4.00 for fun, 3.29 for satisfaction, 3.20 for interest and 3.00 for strangeness. Higher strangeness means feeling more strange, not greater satisfaction. The separate overall observer mean is 3.69. These different perspectives must not be merged into one satisfaction conclusion.

Figure 4 Four user questionnaire items and one observer measure



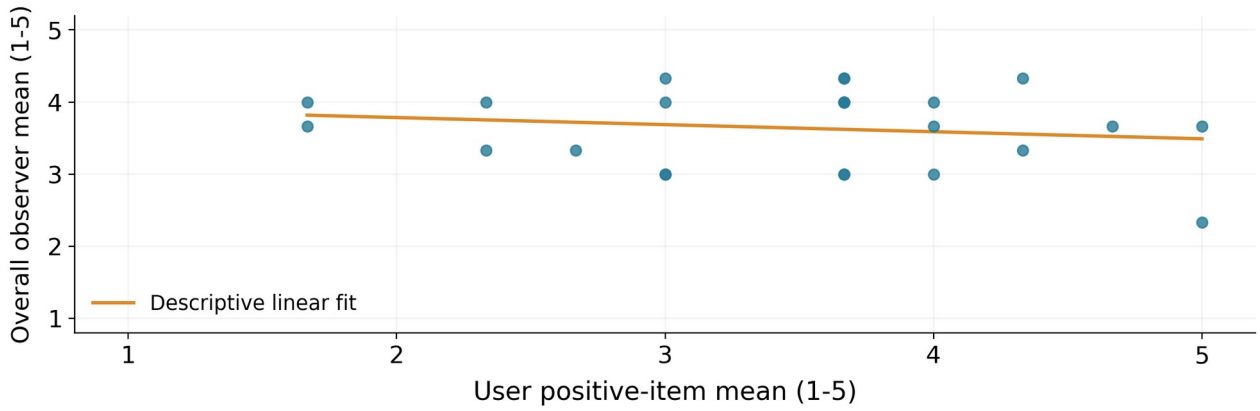
Note. Satisfaction has 24 answers; the other items and overall observer measure have 25 records. The observer measure is not a fifth questionnaire item. Source: Irfan et al. (2024), Overall.

User and observer perspectives

Relationship between user and observer ratings

Each point represents one of 24 complete participants (Figure 5). The horizontal axis shows the user's three-positive-item mean, and the vertical axis shows the overall observer mean. Spearman rank correlation is approximately -0.10, indicating a weak association. This exploratory result does not establish causation or reliable individual prediction. Both perspectives should be considered.

Figure 5 User positive-item mean and overall observer mean

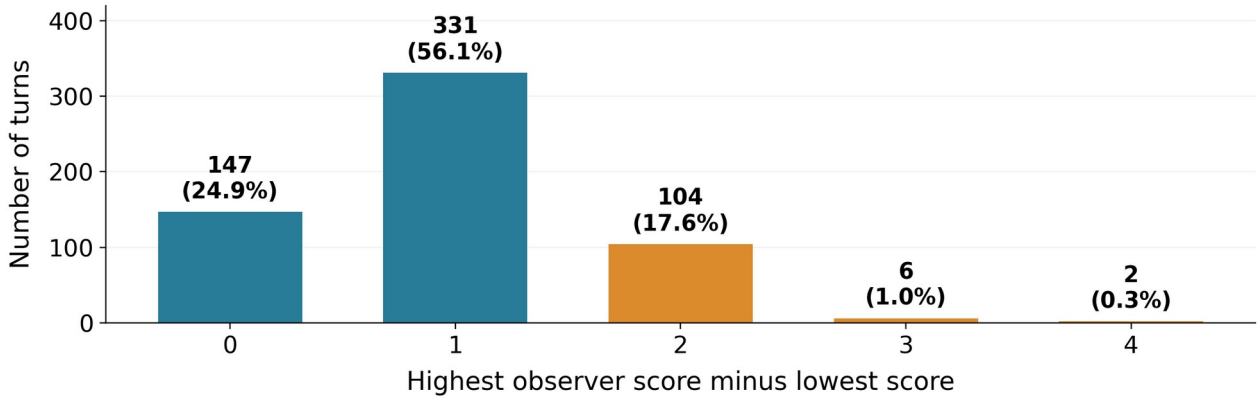


Note. Some points overlap. The orange line is a descriptive linear fit, not a Spearman line. Spearman is calculated separately using ranks. Source: Irfan et al. (2024), Overall.

Score differences between the three observers

Figure 6 counts the score range within each turn: highest minus lowest observer rating. Scores are identical in 147 turns, differ by one point in 331, and differ by at least two in 112. Blue bars therefore cover 478 of 590 turns (81.0%). This describes score spread, not formal inter-rater reliability or accuracy.

Figure 6 Observer score spread within each dialogue turn



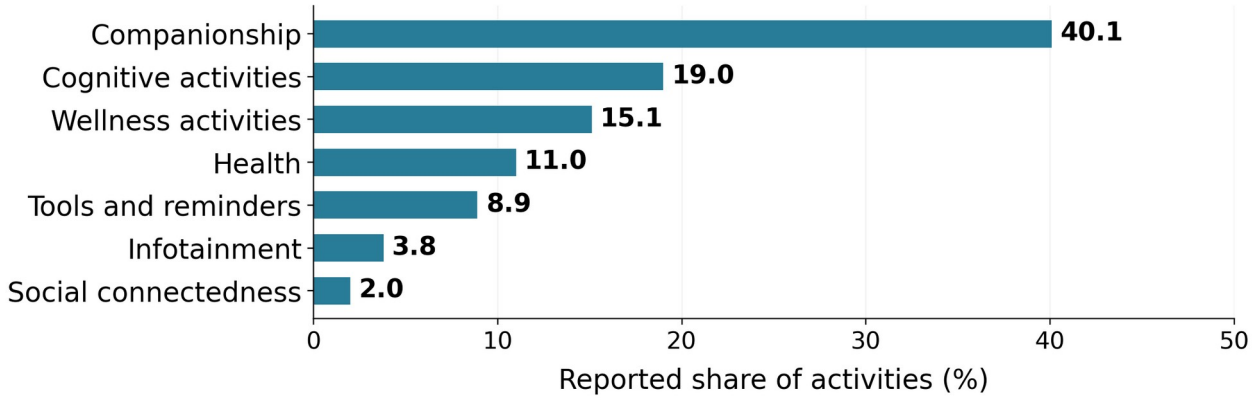
Note. Blue: score range 0-1. Orange: range at least 2. Percentages use all dialogue turns as the denominator. Source: Irfan et al. (2024), Exchanges.

ElliQ activities and feedback

How ElliQ is used in everyday life

HRI CUES describes brief interactions but does not measure long-term use or changes in loneliness. Separate product studies extend these questions to everyday life. In the ElliQ programme report, companionship is the largest activity category at 40.1% (Figure 8). This is a share of activities, not a treatment effect or a percentage of people needing companionship.

Figure 8 ElliQ reported activity categories

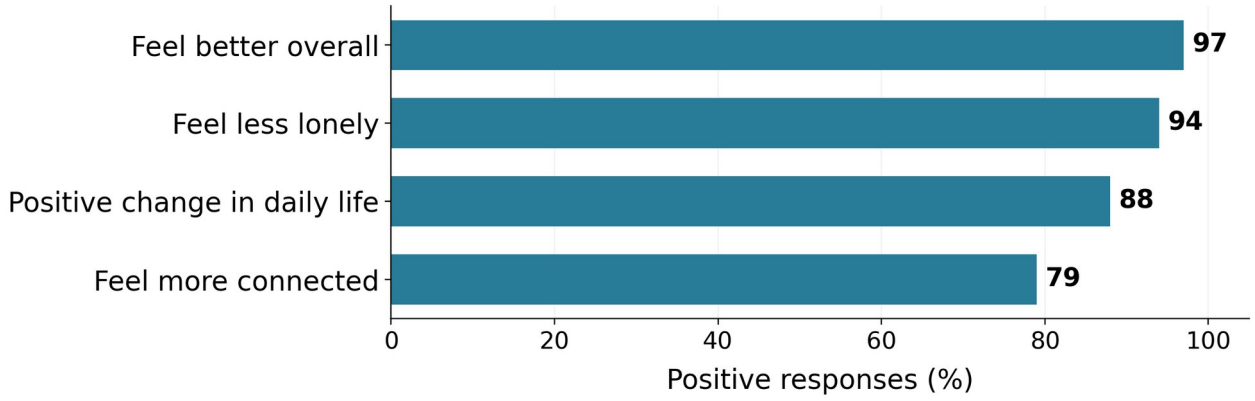


Note. Published shares total 99.9% because of rounding. Source: Intuition Robotics (2026), p. 6.

What users reported about the experience

The same report states that 94% of responding users feel less lonely (Figure 9). This means a share of positive responses, not a 94% fall in loneliness scores. The effective respondent count for each item is not reported. These are programme usage data and self-reports in a manufacturer-authored report, which should be interpreted alongside controlled studies.

Figure 9 ElliQ self-reported feedback



Note. Per-item respondent counts are not provided. Percentages describe positive responses, not changes in loneliness scale scores. Source: Intuition Robotics (2026), p. 3.

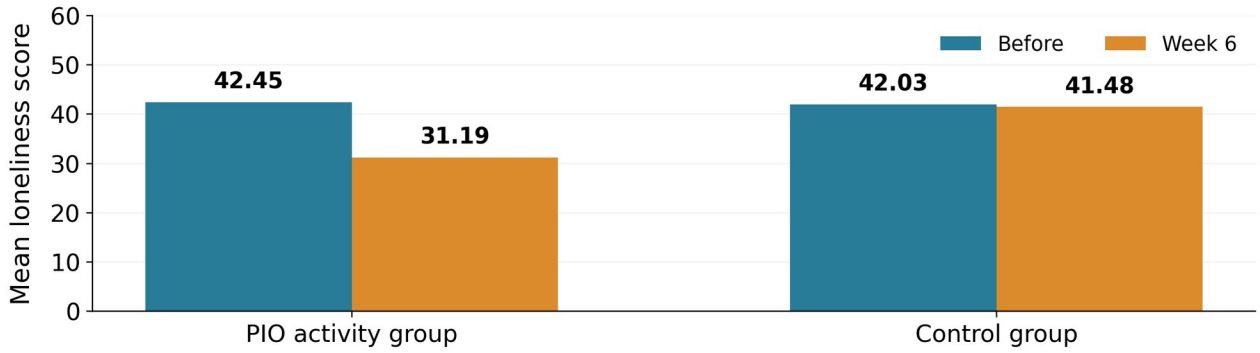
PIO before and after comparison

Loneliness scores before and after six weeks

Lim (2023) studied older adults aged 65 and over living alone in Seoul. Groups were not randomised and continued their usual senior-centre activities. The activity group also joined a staff-led PIO programme twice weekly, 50 minutes per session, for six weeks. Activities included caring for PIO, teaching words, exercise and cognitive games.

Figure 10 compares the group means. The PIO activity group changed from 42.45 to 31.19; the control group changed from 42.03 to 41.48. Lower scores indicate less loneliness.

Figure 10 PIO loneliness scores before and after six weeks

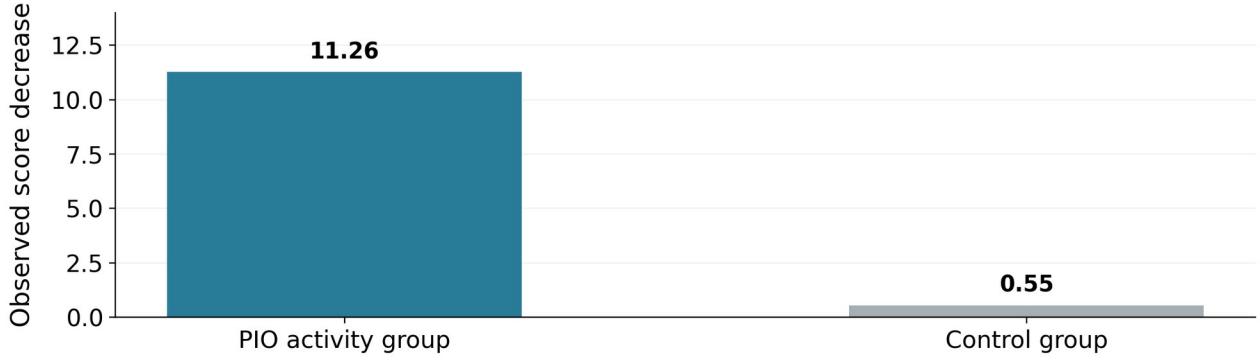


Note. PIO activity group: n=31; control group: n=33. Non-randomised comparison. Source: Lim (2023), Table 4.

Comparing the observed decreases

Figure 11 subtracts Week 6 from the before score. The observed decreases are 11.26 points for the PIO activity group and 0.55 for the control group, a difference of 10.71 points. This compares decreases, not the two final scores. The programme includes staff and group activities, so the robot's separate contribution cannot be isolated.

Figure 11 Observed decrease in the two PIO study groups



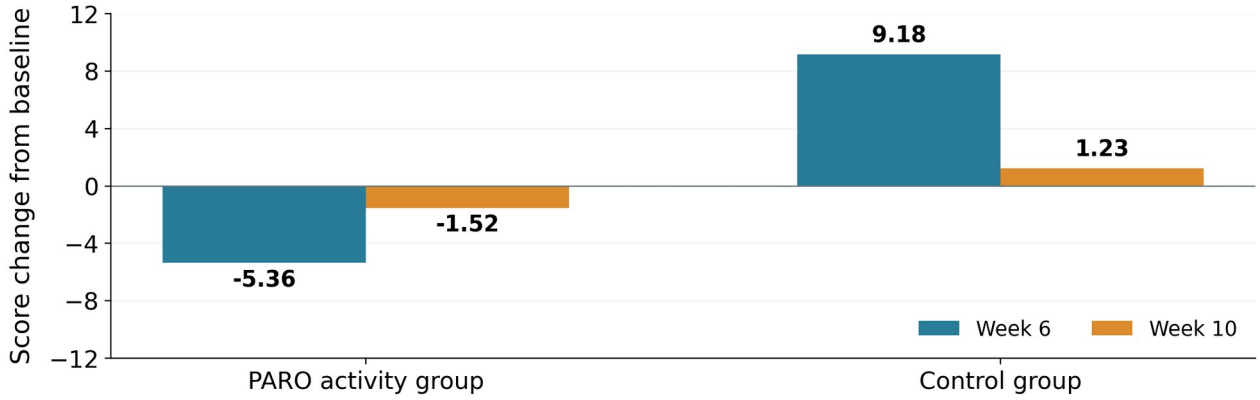
Note. Difference in decreases: 11.26 - 0.55 = 10.71 points. This is descriptive, not a causal effect estimate. Source: Lim (2023), Table 4; calculations in the Notebook.

PARO follow-up and Hyodol retention

PARO loneliness changes at follow-up

In the PARO study, the activity group's mean loneliness score changed by -5.36 points at Week 6 and -1.52 at Week 10 relative to baseline (Figure 12). Negative values indicate lower loneliness. The Week 10 between-group change comparison has p=.07, so a sustained advantage is not established. These results cannot be directly ranked against PIO because the studies differ.

Figure 12 PARO changes in loneliness from baseline

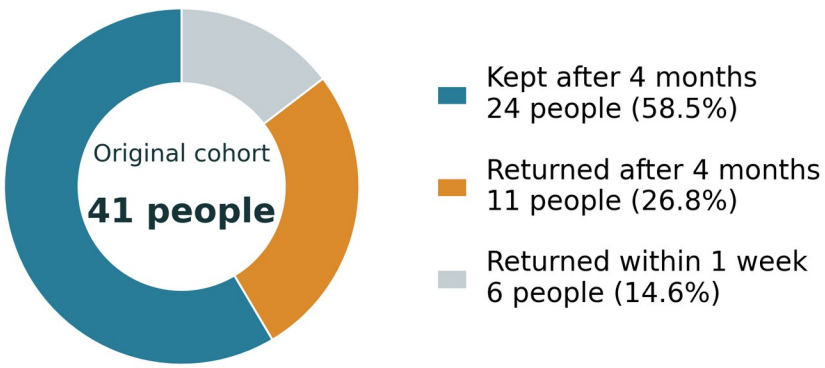


Note. Analysed groups: PARO n=58; control n=60. Values are changes from baseline, not final scores. Source: Chen et al. (2024), Table 4.

Hyodol retention and return after four months

Continued use is a separate question. In the Hyodol study, 24 of the original 41 participants kept the robot after four months (58.5%; Figure 13). Another 11 returned it after four months and 6 within one week. All percentages use the original cohort. Free retention does not establish willingness to buy or prove effectiveness.

Figure 13 Hyodol retention and return in the original cohort

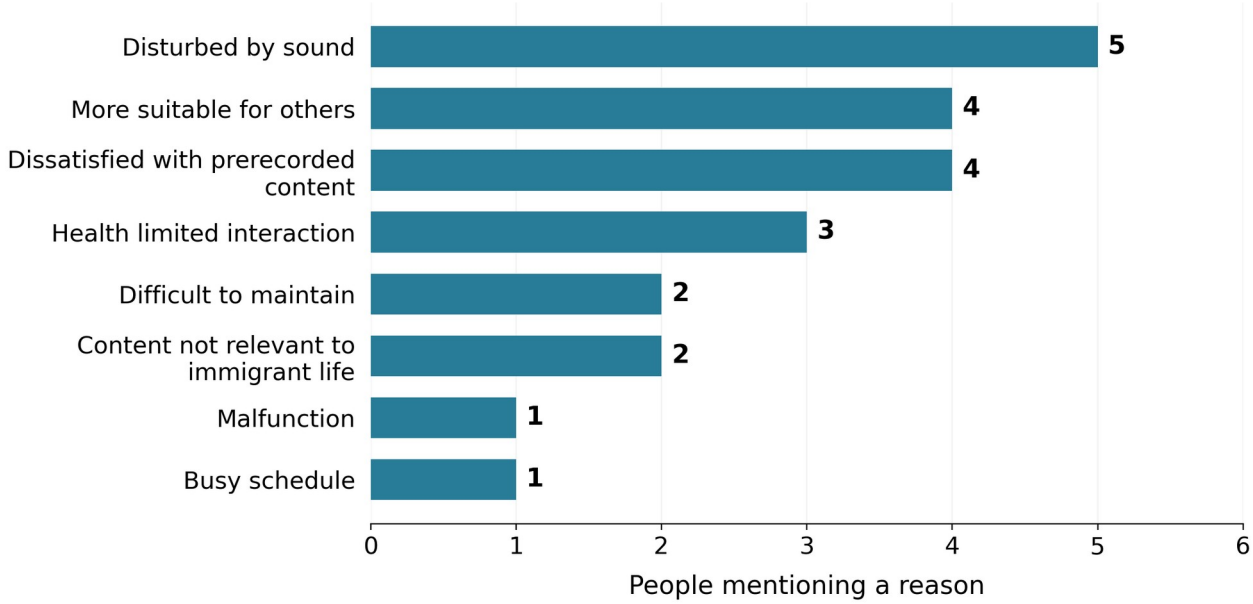


Note. All percentages use the original 41 participants as the denominator. Source: Lee et al. (2024), Methods 2.2 and Results 3.4.

Hyodol return reasons and individual needs

Among the 11 participants who returned Hyodol after four months, five mentioned sound disturbance and four disliked prerecorded content (Figure 14). Participants could give multiple reasons, so the counts must not be added as unique people. The chart contains the eight reported reason counts for this group; blank cells in the original table are not treated as zero.

Figure 14 Reasons for returning Hyodol after four months



Note. Multiple responses from 11 four-month returners. Source: Lee et al. (2024), Table 4.

These reasons suggest testing volume controls, quiet times, topic choices and an easy stop option. The proposed improvements respond to reported difficulties; the study does not show that these changes would increase retention.

Connecting the evidence

HRI CUES highlights differences in individual experience and the need to ask users directly. Hyodol return reasons identify specific settings that could be tested. Together, they support a user-centred design question: can older adults adjust the interaction to fit their needs? New testing is still required.

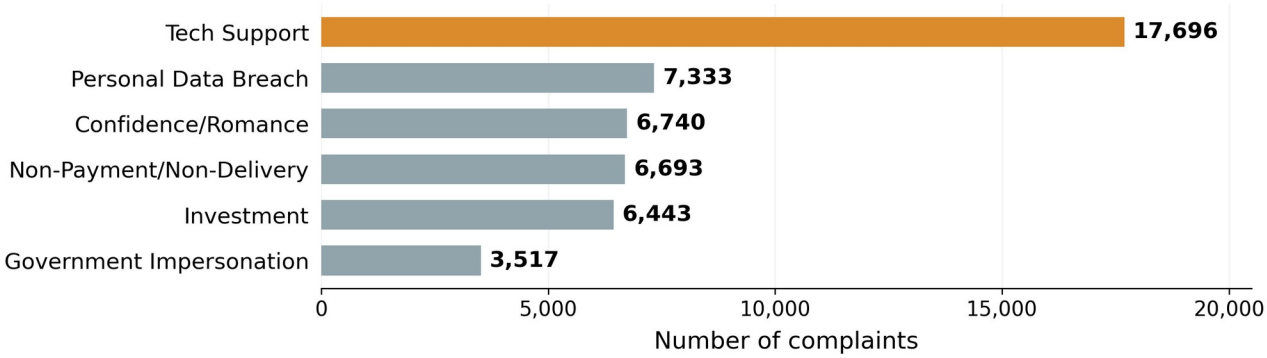
Extending companionship to support functions

Companionship findings do not demonstrate fraud prevention or emergency support. If the product team wants to add these functions, it needs separate evidence about the situation and the reliability of the workflow. The following analysis keeps those two questions distinct.

Fraud context and Hobbit availability

Among six selected IC3 complaint categories for people over 60 in the United States in 2023, Tech Support has the highest count at 17,696 (Figure 16). This motivates a possible test of prompts to pause and verify suspicious installation requests. The figures are not Hong Kong fraud rates, a time trend or evidence that a robot prevents fraud.

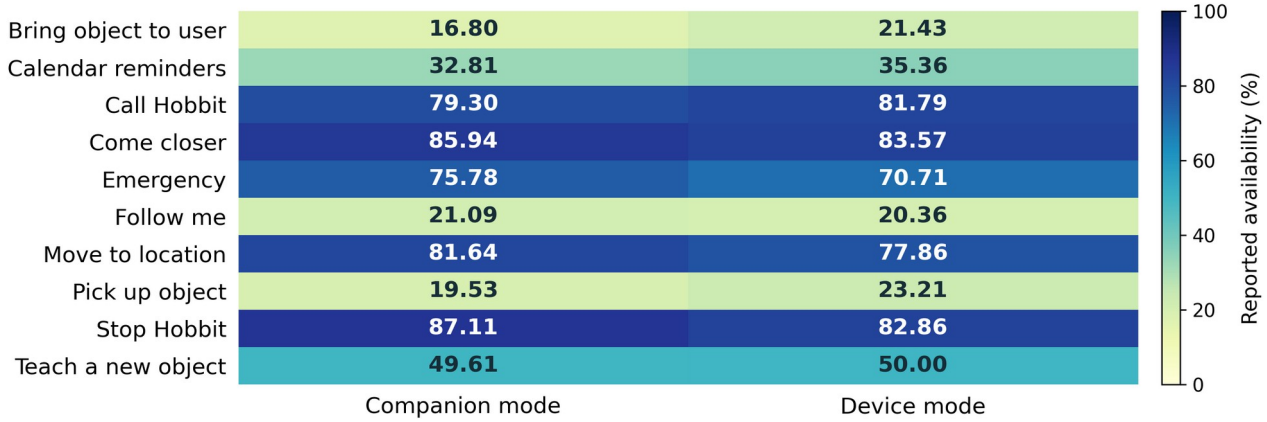
Figure 16 IC3 complaints in six selected categories in 2023



Note. US administrative complaints, age group Over 60. Selected categories, not all complaints. Source: Federal Bureau of Investigation (2024), p. 7.

Hobbit is a separate home-support prototype study. Device mode has Mutual Care disabled. Companion mode adds more social dialogue and proactive interaction. Users experienced Device first, then Companion, so these are sequential conditions rather than randomised parallel groups. The reported emergency-function availability is 70.71% in Device mode and 75.78% in Companion mode. These values describe the authors' Working over days in use indicator, which includes partial working days. They are not detection accuracy, per-task success or rescue rates. The design implication is to test notification confirmation and human fallback, not to copy these percentages as Guardian performance targets.

Figure 15 Hobbit prototype feature availability



Note. Percentages reproduce the authors' availability indicator. Source: Bajones et al. (2018), Table 2. No claim of mode superiority is made.

4 Discussion

The main data identify variation in interaction experience, while the supplementary studies add evidence about use, short-term changes, retention and functional limitations. The sources answer different questions. Their value lies in identifying what to investigate next, rather than combining them into a single product effectiveness claim.

Evidence	Proposed improvement	Records needed
Individual experience varies	Topic choices and brief user feedback	Satisfaction, willingness to use again and reasons for stopping
Sound and prerecorded content can disturb users	Volume, quiet times and a clear stop option	Whether users understand and complete the controls
Suspicious technical requests are a relevant test scenario	Pause and verify with a trusted person	Correct prompts, false alarms and missed prompts in simulations
Prototype function availability varies	Notification confirmation and human fallback	Delivery, delay, acknowledgement and failure records

The proposed improvements are inferences from the evidence, not results of a Guardian trial. The HRI CUES sample is small and comes from Swedish workshop interactions. It does not measure local demand, long-term loneliness or safety performance. Product studies differ in design, sample and outcome measure, and the custom three-positive-item average is exploratory.

The analysis therefore supports a small supervised test before wider use. Any monitoring should be limited to necessary information agreed by the user. The team should define how to confirm a possible problem rather than treat a brief lack of response as an emergency.

Role of the data analyst

The data analyst checks evidence, explains limitations and proposes measurable questions. The research and development team would implement and test the functions with the centre and users. This report does not claim that a working monitoring, fraud detection or emergency system has been built.

5 Conclusion and Guardian proposal

The evidence supports attention to individual experience, suitability for continued use and reliable support workflows. It provides a basis for proposing improvements and tests. It does not establish that Guardian reduces loneliness, prevents fraud or performs successful rescues.

6 Recommendation

Guardian is a proposed companion robot concept. The recommended starting point is to let users choose topics, adjust volume and stop the interaction. A separate support workflow could help a user pause and verify a suspicious request with a trusted person. Both require testing before wider supervised use.

One example of the proposed support workflow

A fictional caller asks an older adult to install unfamiliar software. Guardian displays a short prompt to pause and verify the request. The user can stop, dismiss the prompt or choose to contact a trusted person. If contact is requested, the system checks whether the notification was delivered and acknowledged. If it fails, it explains the failure and offers a staff-supported fallback.

This is a design example, not an automatic judgement that the caller is fraudulent. The prototype should use fictional messages and test contacts. The accompanying Guardian HTML demonstrates preset scenarios only; it does not perform real detection, calls or rescue.

What the proposal needs to demonstrate first

The first questions are practical: can a user understand the prompt, stop the interaction, choose verification and recognise a failed notification? A record that a message was sent is not proof that someone received or acted on it. The proposed test therefore distinguishes sending, delivery and acknowledgement.

A four-week plan is proposed. It involves preparation, staff simulation, a small voluntary interface trial and retesting. The detailed procedure is set out on the next page. None of these tests has been conducted and no new participant data are reported.

Proposed test procedure

The research and development team and the older-adult centre would approve the procedure before inviting volunteers. The purpose is to find usability and workflow problems in a supervised setting. It is not a validation of fraud detection or emergency safety.

Week	Procedure	What to record
1	Agree topics, volume, quiet times, trusted contacts and information sharing. Demonstrate stopping and withdrawing consent.	Chosen settings; whether stopping and withdrawal can be completed.
2	Staff run 20 scripted cases, including suspicious requests, ordinary messages and notification failures. Define the expected response for each case before testing.	Expected versus actual prompt; false alarms; missed prompts; delivery, delay and acknowledgement.
3	Invite five voluntary older adults to try a small set of fictional scenarios with staff present. Ask them to explain the prompt and try stopping or requesting verification.	Understanding; completed actions; help needed; feedback and willingness to use again.
4	Fix problems, repeat the affected cases and review the records with the team and centre.	Whether each issue is resolved on retest; remaining problems and the decision on further supervised use.

Definitions for interpreting the records

In the scripted staff test, a false alarm is a warning in a case predefined as requiring no warning. A missed prompt is the absence of a warning in a case predefined as requiring one. These counts describe the test cases only. A missed acknowledgement or a delivery failure is a separate workflow problem and should be recorded separately.

Decision after the test

Before considering a larger supervised trial, confirm that users can stop and withdraw, that test records are complete, and that known notification failures have been corrected and retested. The team should review remaining problems and agree any further criteria before the next stage. Five volunteers and 20 scripted cases cannot establish population-level effectiveness or product safety.

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Appendix A

Report and Notebook crosswalk

Report section	Notebook figure	Report page
Population background	7	3
Dialogue turns and duration	1, 2	5
Individual experience	3, 4	6
User and observer perspectives	5, 6	7
ElliQ use and feedback	8, 9	8
PIO before and after comparison	10, 11	9
Follow-up and retention	12, 13	10
Return reasons	14	11
Fraud context and availability	16, 15	12

The Notebook retains its English code, Markdown and original figure numbers. The report follows the research argument rather than numerical figure order. Chart labels in this report are in English; data values and definitions follow the checked source files.

The two HRI CUES CSVs remain unchanged. HRI_CUES_Excel_Readable.xlsx is a reading copy, not a replacement for the CSVs used by the Notebook. Swedish source text is preserved and the missing satisfaction answer is not filled in.

The nine supplementary CSVs contain source URLs and table locators. Their 61 rows are aggregate records from different sources, not additional HRI CUES participants. The offline chart HTML provides separate interactive views. The Guardian HTML is a concept demonstration only.

The body includes the summary, introduction, methods, results, discussion, conclusion and recommendations. References and product illustrations are provided separately. Appendix B retains the source report's product and price context as historical reference.

Appendix B Product illustrations

These illustrations identify the products discussed in the supplementary studies. They are not a direct comparison under identical conditions. HRI CUES uses a different conversational configuration. Price information was recorded in the source report as checked on 15 October 2026; it retains the original currencies and is not a quotation for delivery in Hong Kong.

ElliQ desktop device and tablet



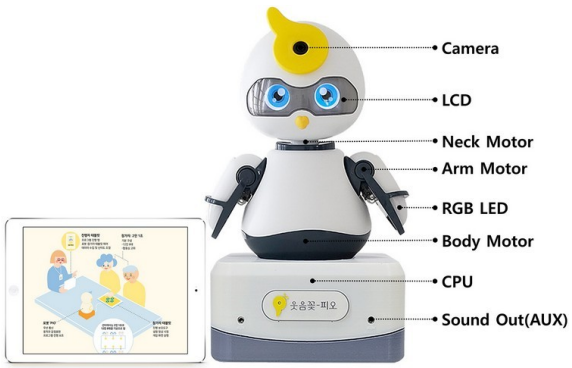
Conversation and everyday companionship activities. See Figures 8 and 9 for activity shares and self-reported feedback.

Price recorded in the source report: US annual membership USD 588 plus a one-time rental initiation fee of USD 249.

Image source: ElliQ official website, https://elliq.com/

Price context: ElliQ membership page, recorded access date 15 October 2026: https://elliq.com/products/membership

PIO parrot shaped social robot



Used in staff-led cognitive and social activities. See Figures 10 and 11 for the six-week loneliness score comparison.

A current public retail price was not verified in the source report. A supplier quotation would be needed.

Image source: Lim (2023), Figure 2. https://doi.org/10.3389/fpubh.2023.1097485

Price context: No verified current public retail quotation was obtained.

Product illustrations continued

PARO seal shaped robot



Used in group interaction and companionship activities. See Figure 12 for loneliness score changes at Weeks 6 and 10.

Historical reference: USD 6,000 in the 2009 US launch announcement, including one year of service and warranty. A current quotation would be needed.

Image source: Aaron Biggs, Paro robot, CC BY-SA 2.0. https://commons.wikimedia.org/wiki/File:Paro_robot.jpg

Price context: PARO Robots US press release, 4 November 2009. https://www.parorobots.com/pdf/pressreleases/PARO%20Robots%20US-Press%20Release%202009-11-04.pdf

Hyodol doll shaped companion robot



A home companion product studied through retention and return experiences. See Figures 13 and 14.

A current public retail price was not verified in the source report. A supplier quotation would be needed.

Image source: Lee et al. (2024), Figure 1. https://doi.org/10.3389/fpubh.2024.1424123

Price context: No verified current public retail quotation was obtained.

Hobbit home support prototype

Hobbit mobile research prototype



A wheeled prototype with a screen and robotic arm for home support research. See Figure 15 for reported feature availability.

Historical development target: a starting price below EUR 14,000. This was a target in the 2011-2015 project, not a current commercial selling price.

Image source: TU Wien ACIN. https://www.acin.tuwien.ac.at/en/vision-for-robotics/roboter/hobbit/

Price context: TU Wien ACIN HOBBIT project page, project period 2011-2015. https://www.acin.tuwien.ac.at/en/project/hobbit/

Device and Companion refer to conditions in the prototype study. Its reported availability percentages do not measure Guardian performance. In this report, the findings are used to motivate notification checks, failure handling and human fallback in a future proposal.